Vedant Desai

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EDUCATION

University of Michigan — Expected May 2028
B.S. in Computer Science and Economics — Ann Arbor, MI

- **GPA:** 3.66 / 4.0
- **Coursework:** Data Structures & Algorithms, Intro to Statistics and Data Analysis, Microeconomics, Macroeconomics, Discrete Mathematics, Calculus III, Physics (Mechanics)

PROJECTS

Granular Synthesizer Plugin | *C++, JUCE* — github.com/Verdent06/granular-synth
Real-time audio DSP plugin built from scratch in C++/JUCE — sine oscillator through full granular engine with lock-free threading and release-ready VST3/AU binaries

- Wired UI-to-audio delivery through a hand-rolled SPSC FIFO (64-slot fixed array, atomic head/tail with acquire/release ordering) so slider changes reach the audio thread without a mutex — and routed WAV handoff via atomic shared_ptr swap so the UI thread never blocks processBlock().
- Enforced the audio thread's zero-allocation constraint by pre-allocating a MemoryPool<Grain, 64> slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so processBlock() never calls new or delete after prepareToPlay() completes.

SignalWeaver | *Python, FastAPI*
Multi-signal financial research platform combining fundamentals, macro indicators, and news sentiment (research assistant; not investment advice)

- Built a linear regression combining fundamental, embedding-based sentiment, and LLM-derived sentiment signals into a composite return-prediction score; validated out-of-sample (3.39% R^2, consistent with the low single-digit R^2 typical of real return-prediction literature) to confirm the model wasn't just fitting noise.
- Served composite financial-research scores through async REST endpoints (FastAPI) wrapping fundamentals, MPNet sentiment, and regression logic, instrumented end-to-end at 9.1s p50 / 15.2s p99 across 90 successful runs on 90 tickers.

EXPERIENCE

CaseStudyPrep.AI — Dec 2025 – May 2026
Software Engineer Co-op (Voice AI) — Remote

- Moved audio processing off the UI thread into a Web Worker with an async stream handoff, reducing main-thread blocking time to under 5ms and keeping the real-time audio visualizer at a smooth 60 FPS during active inference.
- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.

Michigan Data Consulting (MDC) — Jan 2026 – May 2026
Data Engineer — Michigan Campaign Finance Network — Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at ∼2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating ∼800 hours of manual pulls across 400 tracked PACs.
- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.

Vylet | *Python, LangGraph, Docker, Redis, Celery* — May 2026 – Present
Founder — Automated lead-sourcing platform for PE/search-fund; live product generating $1,500 MRR across three paying clients — vyletdata.com

- Launched Vylet, automating a ∼30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Shipped a geography-first discovery mode decoupled from a pre-set vertical, surfacing ∼30 leads/month in niche verticals — custard shops, bowling alleys — structurally invisible to vertical-first discovery.

TECHNICAL SKILLS

Languages .... Python, C++, TypeScript, SQL, HTML/CSS
Frameworks .... FastAPI, Flask, JUCE